

Numerical Methods for ODEs: Exercise Week 6

Problem 1:

Implement the two one-step methods with Butcher tableaus

$$\begin{array}{c|cc} 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \hline & 0 & 1 \end{array} \quad \begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$$

and verify empirically for a suitable test problem that you obtain the expected convergence order.

Problem 2:

Consider the initial value problem

$$y'(t) = \left(1 - \frac{t^2}{2}\right) y(t), \quad t \in (0, 4] \quad (1)$$

$$y(0) = 1. \quad (2)$$

1. Find an analytical solution.
 2. Compare the two methods from Problem 1 - which one performs better? Give an explanation.
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Problem 3:

Repeat your comparison of improved Euler method with constant time step and adaptive time step size selection from last week's exercise for the nonlinear problem

$$x''(t) - \mu (1 - x^2(t)) x'(t) + x(t) = 0 \quad (3)$$

for $t \in (0, 50]$ with $x(0) = 0$ and $x'(0) = 2$ for values $\mu \in \{1/10, 1, 15\}$. Try to explain your results. **Note:** you will have to rewrite this second order equation as a system of two first order equations first.